Complex Solution Sets

Solving equations and reporting complete complex solution sets

Directions

- Report *every* solution, real and non-real. A degree-n equation has n solutions counted with multiplicity, so count yours before you stop.
- Write non-real solutions in standard form $a + bi$. A pair such as $2 + 3i$ and $2 - 3i$ may be written $2 \pm 3i$.
- The methods are mixed on purpose: decide for each problem whether to isolate a square, use the quadratic formula, or factor first.

1. Solve over the complex numbers: $x^2 + 49 = 0$

Answer: ________________

2. Solve over the complex numbers: $3x^2 + 75 = 0$

Answer: ________________

3. Solve over the complex numbers: $x^2 - 4x + 13 = 0$

Answer: ________________

4. Solve over the complex numbers, and list all four solutions: $x^4 - 81 = 0$

Answer: ________________

5. A student solved $x^2 + 16 = 0$ and reported the solution set $\{\, 4i \,\}$.

(a) Give the complete solution set.

Answer: ________________

(b) Explain what is missing and why the equation must have that many solutions.

Answer: ________________

6. Solve over the complex numbers: $2x^2 + 4x + 10 = 0$

Answer: ____________________

7. Solve over the complex numbers: $(x-5)^2 + 9 = 0$

Answer: ____________________

8. Solve over the complex numbers, and list all four solutions: $x^4 + 3x^2 - 4 = 0$

Answer: ____________________

9. Consider the equation $x^2 - 2x + 5 = 0$.

(a) Substitute $x = 1 + 2i$ into the left side and simplify. What value do you get?

Answer: ________________

(b) Using part (a), write the complete solution set of the equation.

Answer: ________________

10. Find all zeros of $f(x) = x^3 - 2x^2 + 9x - 18$. Group the terms in pairs to factor first.

Answer: ________________

11. Solve over the complex numbers: $x^2 + 6x + 34 = 0$

Answer: ________________

12. A quadratic with *real* coefficients and leading coefficient 1 has $4 - i$ as one of its solutions.

(a) Write the quadratic in standard form.

Answer: ____________

(b) Justify why $4 + i$ must be the other solution.

Answer: ____________